Motors and generators

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Distinguish electrical-to-mechanical transfer from mechanical-to-electrical transfer.

SUCCESS CRITERIA - I can...

1. Trace energy through a motor and a generator.
2. Identify magnetic interactions in rotation or induction.
3. Include less useful energy transfers.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “motor” and “generator” to explain your answer. Sentence starters are optional.

motor: a device converting electrical energy to mechanical output

generator: a device converting mechanical input to electrical output

efficiency: useful output divided by input

01 Warm-up: bring back an earlier idea

A consistently drawn magnetic-field map has **3** times as many field lines through the same small perpendicular area at A as at B. What does the denser spacing indicate?

Sentence starter: In my work, I used … to show … . I labeled it … .

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: $\eta = P_{useful}/P_{input}$; $E_{input} = E_{useful} + E_{losses}$. Motor: electrical to mechanical; generator: mechanical to electrical. Count losses and any stored-energy change.

Worked reasoning: An idealized motor receives **20 J** electrically and delivers **12 J** mechanically; **8 J** goes to other transfers. A generator runs the energy conversion in the opposite useful direction.

Your turn: Draw separate motor and generator energy-flow diagrams.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Complete two energy chains: battery → motor → turning axle, and turning axle → generator → lamp. **Label each input** and useful output.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A small classroom car motor receives **100 J** of electrical energy and delivers **60 J** of useful mechanical energy. **Complete the energy-flow diagram**, calculate efficiency and **name the energy transferred** to other forms.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

Read down the left column, then down the right. Use the open space to draw, calculate and explain.

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A motor drives a generator that feeds the motor. **Explain why real losses prevent** the loop from running forever without external input.

Sentence starter: I think … . I can point to …, which shows …, so … .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

An electric field of **6 N/C** points left. **Draw the force direction** on **+4 C** and on **-4 C**, then **compare sizes**. **Draw field** and force arrows separately for each charge.

Word help: electric field = force per positive test charge at a location

Sentence starter: Both … . They differ in … because … .

- Field / first contribution
- Force / second contribution

Draw separate arrows; label quantities and units.

Marzano check - circle: 1 2 3

07 Exit check

Which device converts mechanical input into electrical output?

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

A motor draws the same electrical power in two designs, but one delivers more mechanical power. **Compare efficiencies** and **name a measurement needed**.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3